Finite Infimal Sums of λ-Operator Space Tensor Norms

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Status

This is a partial result for Problem 1 in Chavez-Dominguez, Dimant and Galicer, *Operator space tensor norms*, arXiv:2302.03696v1. It gives a closure theorem producing new λ-operator-space tensor norms, but does not classify such norms.

Source Problem

The source paper concludes with the following problem list. Problem 1 asks for more examples of λ-o.s. tensor norms.

> appeared through this monograph.
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> **Problem 1:** All along the monograph there are results where some hypotheses about local reflexivity are not needed in the case of λ-o.s. tensor norms. It would be welcome to have more examples of λ-o.s. tensor norms, in addition to the typical ones (proj, h and h^t) and the $\odot^\theta$ constructed in Section 2.6.
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> **Problem 2:** We introduce in Section 5 the notions of locally right-accessible tensor norm and locally left-accessible tensor norm, but we do not have any example that justifies that these concepts are weaker than their non-local siblings.
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> **Problem 3:** We consider in Section 6 a weak version of the complete bounded approximation property, namely the W*CBAP. We do not know any example of a space without CBAP satisfying W*CBAP. In view of Proposition 6.10 such a space can not be locally reflexive.
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> **Problem 4:** The theory of α-W*CBAP suggested in Remark 6.9 seems to be an interesting topic for future development.
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> **Problem 5:** In Section 11 it would be interesting to have an example of a locally right-accessible mapping ideal which is not right-accessible. Recall that we show in Example 11.14 that $\mathcal{I}$ is an example for the left version of this question.
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> **Problem 6:** In Proposition 11.12 we see that there is an equivalence between right-accessibility of a mapping ideal and of a tensor norm associated to it. For left-accessibility the result is much weaker and just in one direction. It would be important to know if a statement for the other direction is valid: if the mapping ideal is left-accessible, is it true that the associated tensor norm is (locally) left-accessible?
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> **Problem 7:** With respect to natural tensor norms we have barely begun a classi-

In the source notation, the already listed examples include the projective operator-space tensor norm proj, the Haagerup tensor norm h, its transpose h^t, and the family $\odot^\theta$ constructed in Section 2.6 of the source paper.

Idea of the Proof

The usual Haagerup tensor norm is induced by matrix multiplication $(A, B) \mapsto AB$, while the transposed Haagerup tensor norm is induced by $(A, B) \mapsto BA$. Instead of averaging these two products, as in the source's $\odot^\theta$ family, put both products in orthogonal diagonal blocks:

$$(A, B) \longmapsto \begin{pmatrix} AB & 0 \\ 0 & BA \end{pmatrix}.$$

The same idea works for any finite family of λ-products. Put their matrix products in diagonal blocks. The λ-axioms are inherited by compressing to one block for (E1) and by permuting blocks for (E2). On the dual side, the bilinear norm becomes the maximum of the original dual bilinear norms, which is exactly the dual norm of the operator-space infimal sum. Thus finite infimal sums of λ-o.s. tensor norms are again λ-o.s. tensor norms.

The Result

Theorem 1 (Finite infimal-sum closure)**.** *Let $N \geq 1$, and let $\lambda^1, \dots, \lambda^N$ be λ-o.s. tensor norms in the sense of the source paper. Write*

$$B_k^i : M_k \times M_k \longrightarrow M_{\tau_i(k)}$$

for the bilinear maps defining λ^i. Define

$$\nu_k(A, B) = \operatorname{diag}\left(B_k^1(A, B), \dots, B_k^N(A, B)\right) \in M_{\tau_1(k)+\cdots+\tau_N(k)}.$$

Then $\nu = (\nu_k)$ is a λ-o.s. tensor norm. Moreover, for every pair of operator spaces E, F, the induced operator-space tensor norm is the infimal sum

$$\lambda^1 + \cdots + \lambda^N,$$

meaning that at each matrix level

$$\|u\|_{\nu,n} = \inf\left\{\sum_{i=1}^N \|u_i\|_{\lambda^i,n} : u = u_1 + \cdots + u_N\right\}.$$

Proof. We verify the source paper's conditions (E1), (E2), and (E3).

For (E3), since each λ^i satisfies (E3),

$$\nu_1(1, 1) = \operatorname{diag}(B_1^1(1, 1), \dots, B_1^N(1, 1))$$

is the identity matrix of the codomain, and

$$\|\nu_k(A, B)\| = \max_i \|B_k^i(A, B)\| \leq \left(\max_i \|B_k^i\|\right)\|A\|\,\|B\|.$$

Hence $\sup_k \|\nu_k\| < \infty$.

For (E1), fix k. Use the (E1) witnesses for λ^1: some p, matrices S_1, T_1, and matrices $a_1, \dots, a_k \in M_p$. Let

$$S = \begin{pmatrix} S_1 & 0 & \cdots & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} T_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

with zeros in the remaining block columns and rows. Then

$$S\nu_p(a_{j_1}, a_{j_2})T = S_1 B_p^1(a_{j_1}, a_{j_2})T_1,$$

which is the required diagonal matrix unit when $j_1 = j_2$ and zero otherwise. Thus ν satisfies (E1).

For (E2), fix r, s. For each i, let P_i be an (E2) matrix for λ^i. The block-diagonal matrix

$$Q = \operatorname{diag}(P_1, \ldots, P_N)$$

has norm at most one and sends ν_{r+s} to the block diagonal matrix

$$\operatorname{diag}\left(B_r^1, B_s^1, B_r^2, B_s^2, \ldots, B_r^N, B_s^N\right)$$

on the matrix-unit pairs appearing in (E2). A permutation unitary reorders these blocks as

$$\operatorname{diag}\left(B_r^1, \ldots, B_r^N, B_s^1, \ldots, B_s^N\right) = \operatorname{diag}(\nu_r, \nu_s).$$

Multiplying Q by this unitary preserves the norm bound, giving (E2).

It remains to identify the induced tensor norm. Let $\Phi : E \times F \to M_m$ be a completely bounded bilinear map. For every k and $x \in M_k(E)$, $y \in M_k(F)$,

$$\Phi_{\nu_k}(x, y) = \operatorname{diag}\left(\Phi_{B_k^1}(x, y), \ldots, \Phi_{B_k^N}(x, y)\right).$$

Therefore

$$\|\Phi\|_{\mathrm{cb},\nu} = \max_{1 \le i \le N} \|\Phi\|_{\mathrm{cb},\lambda^i}$$

at every matrix level. By the source paper's duality theorem for λ-tensor products, the dual of $E \otimes_\nu F$ is the intersection of the duals of $E \otimes_{\lambda^i} F$, equipped with the maximum of the dual norms.

The dual norm of the matrix-level infimal sum

$$\inf\left\{\sum_i \|u_i\|_{\lambda^i,n} : u = \sum_i u_i\right\}$$

is exactly the maximum of the corresponding dual norms. Hence the matrix-level norms induced by ν agree with the operator-space infimal sum of the λ^i's. Since finite infimal sums of operator-space tensor norms are again operator-space tensor norms, ν is a λ-o.s. tensor norm. □

Theorem 2 (Symmetrized Haagerup corollary)**.** *Let $\nu = (\nu_k)_{k \ge 1}$ be the sequence of bilinear maps*

$$\nu_k : M_k \times M_k \longrightarrow M_{2k}, \qquad \nu_k(A, B) = \begin{pmatrix} AB & 0 \\ 0 & BA \end{pmatrix}.$$

Then ν defines a λ-o.s. tensor norm. For every pair of operator spaces E, F, the induced tensor norm on $E \otimes F$ is completely isometric to the symmetrized Haagerup sum $h + h^t$.

Proof. Apply the finite infimal-sum closure theorem to the λ-products defined by AB and BA, which induce h and h^t, respectively. □

Verification and Limitations

The proof is purely formal from the definitions in Section 2.6 of the source paper and from its duality theorem for λ-tensor products. No computation was used.

The result gives a systematic finite-sum closure construction and, in particular, the concrete additional λ-o.s. tensor norm $h + h^t$. It does not address the existence of spaces with W*CBAP but without CBAP, the accessibility questions, the full natural-norm classification, or the duality identity $(\backslash \min /)' = /\mathrm{proj}\backslash$.

Novelty Check

I searched the run indexes for arXiv:2302.03696 and for the phrases λ-o.s. tensor norm, $h + h^t$, symmetrized Haagerup, infimal sum, block diagonal, and operator space tensor norms. I also searched the web for exact phrases including `"h+h^t" "lambda" "operator space" tensor norm`, `"symmetrized Haagerup" "lambda" tensor product`, `"lambda tensor product" "operator spaces" "direct sum"`, and `"operator space" "lambda tensor norm" "block diagonal"`. No direct prior statement of this closure result was found in the searched material.

Human Review Recommendation

Review the convention in condition (E3), where 1 denotes the identity of the codomain matrix algebra, and verify that the block-permutation argument for (E2) matches the source paper's indexing. The main mathematical mechanism is the dual identification of the block-diagonal product with the matrix-level infimal sum of the original tensor norms.

References

[1] J. Alejandro Chavez-Dominguez, Veronica Dimant, and Daniel Galicer, *Operator space tensor norms*, arXiv:2302.03696v1, 2023.